

Hermite–Biehler Stability from the Anti-Symmetric Pandrosion Quotient

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Abstract

We show that the classical Hermite–Biehler theorem on Hurwitz stability (all zeros in the open left half-plane) admits a natural Pandrosion formulation: $P(z) = P_+(z) + z \cdot Q(-z, z)$, where $Q(-z, z) = (P(z) - P(-z))/(2z)$ is the *anti-symmetric Pandrosion quotient*. Hurwitz stability is equivalent to the interlacing of the zeros of the even part P_+ and the anti-symmetric quotient $Q(-\cdot, \cdot)$ on the imaginary axis. Combined with the Cayley transform, this links the Schur–Cohn criterion (Paper 19, anchor $a = 0$, unit disk) to the Hurwitz criterion (anchor $a = -1$, left half-plane), giving a unified Pandrosion theory of stability.

1 The anti-symmetric Pandrosion quotient

Definition 1.1 (Anti-symmetric Pandrosion quotient). For a polynomial P with real coefficients, the **anti-symmetric Pandrosion quotient** is:

$$Q(-z, z) := \frac{P(z) - P(-z)}{2z}. \quad (1)$$

This is the Pandrosion quotient $Q(a, z) = (P(z) - P(a))/(z - a)$ evaluated at $a = -z$, then divided by 2.

Proposition 1.2 (Even–odd decomposition). For $P(z) = \sum_{k=0}^n a_k z^k$ with real coefficients:

$$\boxed{P(z) = P_+(z) + z \cdot Q(-z, z)}, \quad (2)$$

where

$$P_+(z) := \frac{P(z) + P(-z)}{2} = \sum_{\ell} a_{2\ell} z^{2\ell}, \quad (3)$$

$$Q(-z, z) = \frac{P(z) - P(-z)}{2z} = \sum_{\ell} a_{2\ell+1} z^{2\ell}. \quad (4)$$

P_+ collects the even-degree coefficients; $Q(-z, z)$ collects the odd-degree coefficients (shifted down by one power of z).

Proof. $P(z) = \frac{P(z)+P(-z)}{2} + \frac{P(z)-P(-z)}{2} = P_+(z) + z \cdot \frac{P(z)-P(-z)}{2z} = P_+(z) + z \cdot Q(-z, z)$. $\square$

2 Hermite–Biehler in Pandrosion form

On the imaginary axis $z = i\omega$ ($\omega \in \mathbb{R}$), the decomposition gives:

$$\operatorname{Re}(P(i\omega)) = P_+(i\omega) = \sum_{\ell} (-1)^{\ell} a_{2\ell} \omega^{2\ell}, \quad (5)$$

$$\operatorname{Im}(P(i\omega)) = \omega \cdot Q(-i\omega, i\omega) = \sum_{\ell} (-1)^{\ell} a_{2\ell+1} \omega^{2\ell+1}. \quad (6)$$

The real part is controlled by P_+ (even coefficients), and the imaginary part by $Q(-z, z)$ (odd coefficients).

Theorem 2.1 (Hermite–Biehler, Pandrosion form). *A polynomial P with real coefficients and positive leading coefficient is **Hurwitz stable** (all zeros satisfy $\operatorname{Re}(z) < 0$) if and only if the real zeros of:*

- $P_+(i\omega) = \sum (-1)^\ell a_{2\ell} \omega^{2\ell}$ (the even response), and
- $Q(-i\omega, i\omega) = \sum (-1)^\ell a_{2\ell+1} \omega^{2\ell}$ (the anti-symmetric Pandrosion response)

are all real, simple, and **strictly interlace** on the ω -axis.

Remark 2.2. The classical formulation uses “even part” and “odd part”. The Pandrosion perspective replaces the odd part by $Q(-z, z) = (P(z) - P(-z))/(2z)$, which is the value of the difference quotient at the *reflected anchor* $a = -z$. This is geometrically natural: instability corresponds to the failure of the “reflection quotient” to interlace properly with the even part.

3 Examples

$P(z)$	Roots	Re/Im zeros (ω)	Interlace?
$z^3 + 2z^2 + 3z + 4$	$-1.65, -0.17 \pm 1.55i$	$\pm 1.41 / 0, \pm 1.73$	✓ Hurwitz
$z^3 + z^2 + z + 4$	$-1.74, 0.37 \pm 1.47i$	$\pm 2.0 / 0, \pm 1.0$	× Not Hurwitz

In the Hurwitz case, the pattern on the ω -axis is I-R-I-R-I (alternating). In the non-Hurwitz case, the pattern is R-I-I-I-R (non-alternating).

4 Link to Schur–Cohn via Cayley transform

Theorem 4.1 (Cayley bridge). *The Cayley transform $w = (z - 1)/(z + 1)$ maps the unit disk $|z| < 1$ to the left half-plane $\operatorname{Re}(w) < 0$. Under this map:*

- Schur $\rightarrow$ Hurwitz:** If $P(z)$ has all zeros in $|z| < 1$, then $\tilde{P}(w) = (w + 1)^d P\left(\frac{1+w}{1-w}\right)$ has all zeros in $\operatorname{Re}(w) < 0$.
- Anchor correspondence:** The Pandrosion anchor $a = 0$ (Schur, Paper 19) maps to $a = -1$ (Hurwitz):

$$Q_{\text{Schur}}(0, z) = \frac{P(z) - P(0)}{z} \quad [\text{disk stability}], \quad (7)$$

$$Q_{\text{Hurwitz}}(-1, w) = \frac{\tilde{P}(w) - \tilde{P}(-1)}{w + 1} \quad [\text{half-plane stability}]. \quad (8)$$

Criterion	Region	Anchor	Pandrosion object	Paper
Schur–Cohn	$ z < 1$	$a = 0$	$T = \bar{a}_n Q(0, z) - a_0 Q^*(0, z)$	19
Hermite–Biehler	$\operatorname{Re}(z) < 0$	$a = -z$	$P = P_+ + z \cdot Q(-z, z)$	24
Eneström–Kakeya	$ z \leq 1$	$a = 1$	$Q(1, z)$ duality	16
Gauss–Lucas	conv	$a = \alpha_k$	$\sum Q(\alpha_k, z) = 0$	14

5 Conclusion

The Hermite–Biehler theorem is the *half-plane analogue* of the Schur–Cohn criterion (Paper 19), with the Pandrosion anchor moving from $a = 0$ (unit disk) to $a = -z$ (left half-plane, via reflection). The anti-symmetric Pandrosion quotient $Q(-z, z)$ extracts the odd part of P , and stability is encoded in the interlacing of even and odd responses on the imaginary axis.

The four stability/containment criteria of the series—Gauss–Lucas (anchor α_k), Eneström–Kakeya (anchor 1), Schur–Cohn (anchor 0), and Hermite–Biehler (anchor $-z$)—are all instances of a single principle: *polynomial zero localization is controlled by the structure of the Pandrosion quotient $Q(a, z)$ at appropriate anchors.*

References

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